



Karunya INSTITUTE OF TECHNOLOGY AND SCIENCES

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Division of Electronics and Communication Engineering
III IA EVALUATION REPORT
for
BLENDED LEARNING PROJECT-BASED LEARNING

**“Parity-Based Error Detection System for
Digital Communication Systems”**

A report submitted by

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Register Number	URK23EC4001, URK23EC4012, URK23EC4017, URK23EC4029
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Project Rubrics for Evaluation

First Review: to be awarded for **10 Marks** – PPT should have four slides (Title page, Introduction, Circuit/Block Diagram, and Description of Project).

Second Review: to be awarded for **10 Marks** – PPT should have three slides (Description of Circuit/Concept, Design of Circuit in Tools/Design of Flow graph, and Partial Results)

Third Review: to be awarded for **10 Marks** – PPT should have two slides (Tool based simulation/Hardware based simulation/Concept based interaction/ Case Study)

Fourth Review: to be awarded for **10 Marks** –PPT should have two slides (Output Results & Discussion)

[NOTE: The instructions are subjected to change if required to combine either of two reviews and to be evaluated for 20 Marks.]

Total marks: _____ / 40 Marks

Signature of Faculty with date:



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Karunya Nagar, Coimbatore - 641 114, Tamil Nadu, India.

BONAFIDE CERTIFICATE

This is to certify that the project report entitled,” Parity-Based Error Detection System for Digital Communication Systems” is a bonafide record of the project work carried out by the following candidate for the course 23EC2007-VLSI design during the academic year 2025-2026.

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ABSTRACT

The advancement of digital communication systems has increased the need for reliable data transmission. During communication, data can be corrupted due to noise, interference, or hardware limitations. To address this issue, error detection techniques are widely used to ensure data integrity. This project presents the design and implementation of a 4-bit parity-based error detection system for digital communication applications.

The proposed system is designed using digital logic circuits and implemented in Cadence Virtuoso, which serves as the design and simulation platform. The system consists of two main components: a parity generator and a parity checker. The parity generator produces a parity bit for a 4-bit input data sequence using XOR logic, ensuring that the transmitted data follows either even or odd parity.

At the receiver side, the parity checker recalculates the parity using the received data bits along with the transmitted parity bit. By comparing the result, the system determines whether an error has occurred during transmission. If the parity condition is violated, the system indicates the presence of an error; otherwise, it confirms that the data is error-free.

The entire design is implemented at the schematic level using basic logic gates such as XOR gates in the Cadence environment. Simulation is carried out to verify the correctness of the circuit for different input combinations. The results demonstrate that the system successfully detects single-bit errors in transmitted data.

Overall, this project highlights a simple yet effective method for error detection in digital systems. It demonstrates the practical application of parity checking techniques and provides a foundation for more advanced error detection and correction methods in communication systems.



TABLE OF CONTENTS

CHAPTER	TITLE	PAGE NO.
1.	INTRODUCTION	3
2.	PROBLEM STATEMENT	4
3.	AIM	5
4.	THEORY	6
5.	METHODOLOGY	7
6.	OUTPUT	10
7.	CONCLUSION	11

CHAPTER 1 INTRODUCTION

In modern digital communication systems, the accurate transmission of data is of utmost importance. However, during data transfer, signals are often affected by noise, interference, and hardware imperfections, which may lead to errors in the received data. Even a single-bit error can significantly impact the performance and reliability of a system. Therefore, error detection techniques play a crucial role in ensuring data integrity.

One of the simplest and most widely used error detection methods is parity checking. In this technique, an additional bit known as the parity bit is appended to the data before transmission. This bit is generated based on the number of logic high (1's) present in the data. Depending on the type of parity used (even or odd), the system ensures that the total number of 1's follows a specific rule. At the receiver side, the parity is recalculated and compared with the received parity bit to determine whether an error has occurred.

This project focuses on the design and implementation of a 4-bit parity-based error detection system using digital logic circuits. The system consists of a parity generator, which generates the parity bit for a 4-bit input data, and a parity checker, which verifies the correctness of the received data. The design is implemented using basic logic gates, primarily XOR gates, due to their suitability for parity operations.

The entire system is designed and simulated using Cadence Virtuoso, a widely used tool for VLSI design and circuit simulation. Through schematic design and simulation, the functionality of the parity generator and checker is verified for various input combinations.

This project provides a clear understanding of fundamental error detection techniques and demonstrates their practical implementation in digital systems. It also lays the groundwork for more advanced techniques such as error correction codes and reliable communication system design.

CHAPTER 2 PROBLEM STATEMENT

In digital communication systems, data is transmitted from a sender to a receiver through various communication channels. During this transmission, the data is highly susceptible to errors caused by noise, interference, signal attenuation, and hardware imperfections. These errors can alter one or more bits of the original data, leading to incorrect information being received and processed.

Without an efficient error detection mechanism, it becomes difficult to identify whether the received data is accurate or corrupted. This can result in system failures, unreliable communication, and incorrect outputs in critical applications such as computing systems, embedded devices, and data transmission networks.

Therefore, there is a need for a simple, efficient, and reliable method to detect errors in transmitted data. The challenge is to design a system that can identify errors with minimal hardware complexity and high speed.

This project addresses the problem by designing and implementing a 4-bit parity-based error detection system using digital logic circuits in Cadence. The system aims to detect errors in transmitted data by using parity checking techniques, thereby improving the reliability of digital communication systems.

CHAPTER 3 AIM

The aim of this project is to design, implement, simulate, and compare a 4-bit parity-based error detection system for digital communication applications using Cadence Virtuoso, with a detailed analysis of power consumption in parity generator and parity checker circuits designed using Static CMOS XOR logic and Transmission Gate XOR logic. The project is intended to develop a reliable system that detects errors occurring during data transmission by generating parity bits for 4-bit input data at the transmitter side and verifying them at the receiver side.

In this work, both the parity generator and parity checker are constructed using XOR-based digital logic circuits, since XOR gates play a vital role in parity computation. Two different XOR design techniques—Static CMOS and Transmission Gate logic—are implemented separately to evaluate their effectiveness in circuit performance. The comparison focuses mainly on power dissipation, while also considering circuit reliability and suitability for low-power VLSI communication systems.

Through schematic design, simulation, and performance evaluation in Cadence Virtuoso, the project aims to identify the most power-efficient XOR implementation method for parity-based error detection. This study not only demonstrates the practical application of parity checking in maintaining data integrity in communication systems but also provides insight into optimized low-power digital circuit design for future advanced error detection and correction technologies.

CHAPTER 4 THEORY

In digital communication systems, data is transmitted in the form of binary bits (0s and 1s). During transmission, the data may get corrupted due to noise, interference, or other disturbances present in the communication channel. To ensure the accuracy and reliability of transmitted data, error detection techniques are employed. One of the simplest and most commonly used methods for error detection is parity checking.

4.1 Parity Concept

Parity is an error detection technique in which an additional bit, known as the parity bit, is appended to the original data before transmission. This bit is generated such that the total number of logic '1's in the data follows a predefined rule.

There are two types of parity:

- Even Parity: The total number of 1's (including the parity bit) is even.
- Odd Parity: The total number of 1's (including the parity bit) is odd.

Parity checking helps in identifying whether the transmitted data has been altered during communication.

4.2 Parity Generator

A parity generator is a combinational logic circuit used to generate the parity bit for a given set of data bits. In this project, a 4-bit data input (A, B, C, D) is considered.

The parity bit is generated using XOR (Exclusive-OR) logic as follows:

$$P = A \oplus B \oplus C \oplus D$$

The XOR operation produces a logic '1' when the number of input 1's is odd, making it suitable for parity generation.

4.3 Parity Checker

At the receiver end, a parity checker is used to verify the correctness of the received data. It recalculates the parity by performing XOR operation on all received data bits along with the transmitted parity bit.

$$E = A \oplus B \oplus C \oplus D \oplus P$$

Where:

- $E = 0$ indicates that no error is detected
- $E = 1$ indicates that an error is detected

4.4 Working Principle

The working of the parity-based error detection system can be explained as follows:

1. The sender generates a parity bit using the input data bits.
2. The data along with the parity bit is transmitted through the communication channel.
3. At the receiver, the parity is recalculated using the received bits.
4. The recalculated parity is compared with the received parity bit.
5. If both match, the data is considered error-free; otherwise, an error is detected.

4.5 Advantages

- Simple and easy to implement
- Requires minimal hardware
- Provides fast error detection
- Suitable for basic communication systems

4.6 Limitations

- Can effectively detect only single-bit errors
- Cannot detect errors if an even number of bits are corrupted
- Does not support error correction, only error detection

CHAPTER 5 METHODOLOGY

The methodology of the proposed 4-bit parity-based error detection system involves the design, implementation, and simulation of parity generator and parity checker circuits using digital logic in Cadence Virtuoso. The overall process is carried out in a systematic manner as described below:

5.1 System Design

The system is divided into two main parts:

Parity Generator (Transmitter Side)

Parity Checker (Receiver Side)

The parity generator is responsible for generating the parity bit for the input data, while the parity checker verifies the correctness of the received data.

5.2 Design of Parity Generator

Four input bits (A, B, C, D) are provided as input.

XOR gates are used to generate the parity bit.

The inputs are connected in a cascaded manner using XOR logic.

The output obtained is the parity bit (P), which is transmitted along with the data.

5.3 Design of Parity Checker

The receiver accepts four data bits along with the parity bit.

All inputs (A, B, C, D, P) are given to XOR gates.

The output of the XOR operation indicates the presence or absence of error. If the output is '0', the data is correct; if it is '1', an error is detected.

5.4 Cadence Implementation

The complete circuit is implemented using Cadence Virtuoso Schematic Editor with the following steps:

- Create a new library and cell view.
- Select appropriate technology library (e.g., gpd180).
- Design XOR gates using standard cells or basic logic gates.
- Construct the parity generator circuit using XOR gates.
- Design the parity checker circuit similarly.
- Connect all components properly and label inputs/outputs.
- Perform schematic verification to ensure correctness.

5.5 Simulation and Testing

The designed circuits are simulated using Cadence simulation tools (such as Spectre).

Different combinations of input data are applied.

The generated parity bit is verified.

Errors are intentionally introduced to test the parity checker. The output is observed to confirm correct error detection.

5.6Result Verification

The system is verified for all possible input combinations.

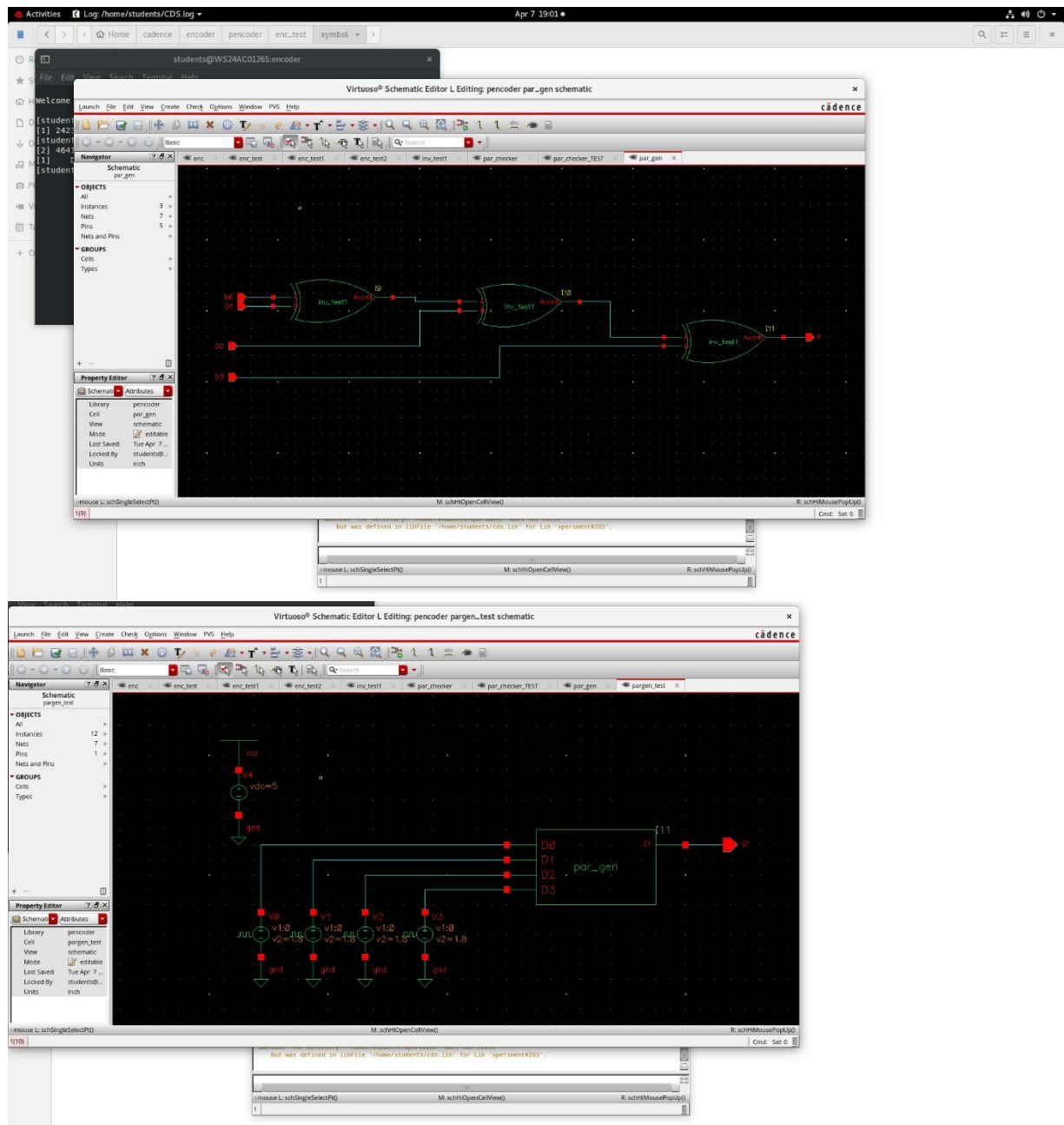
The parity generator correctly produces the parity bit.

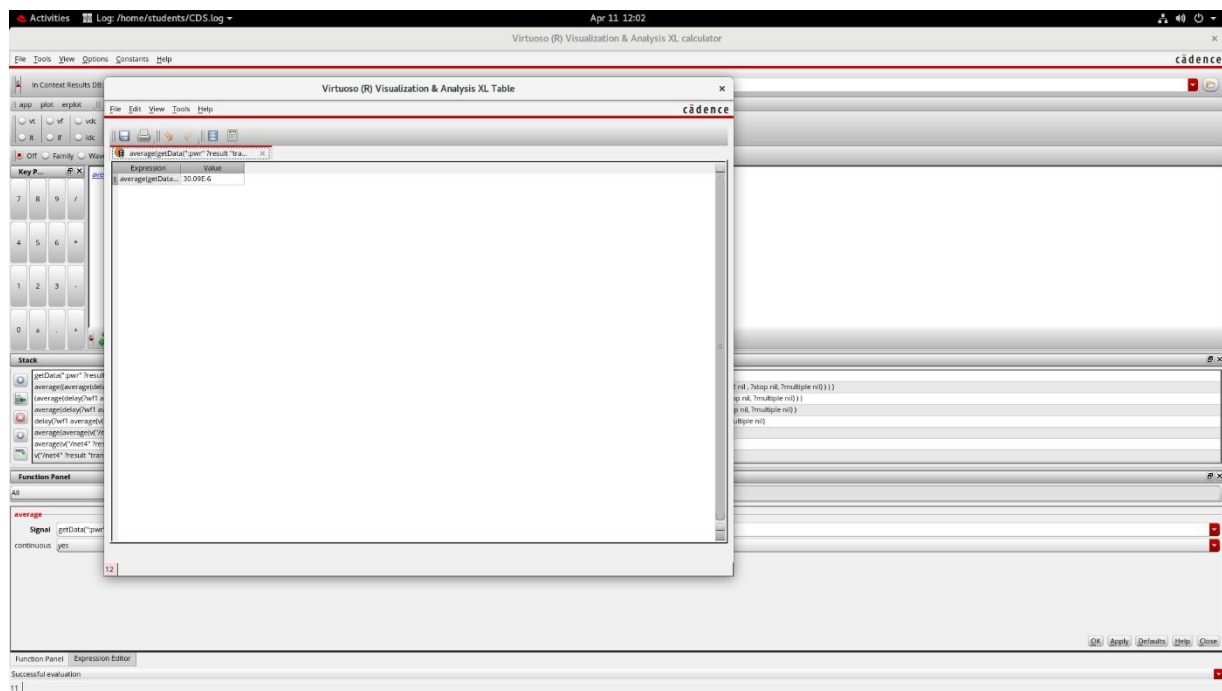
The parity checker successfully detects errors when present.

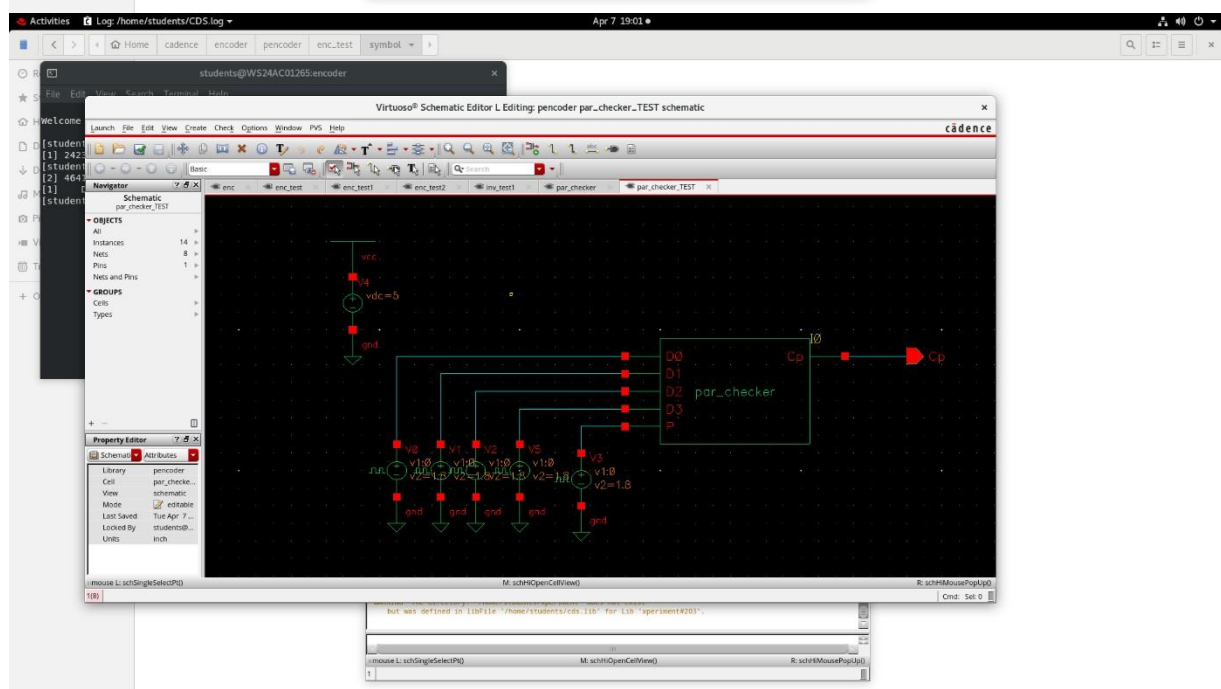
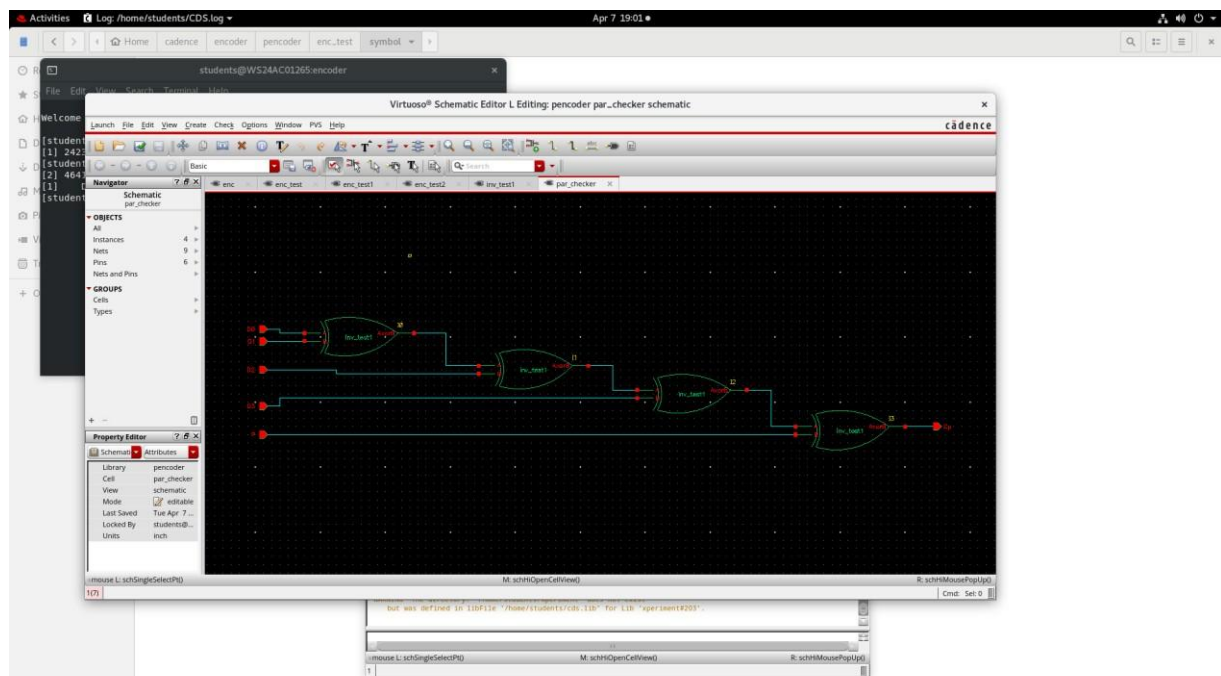
Simulation results confirm the proper functioning of the system.

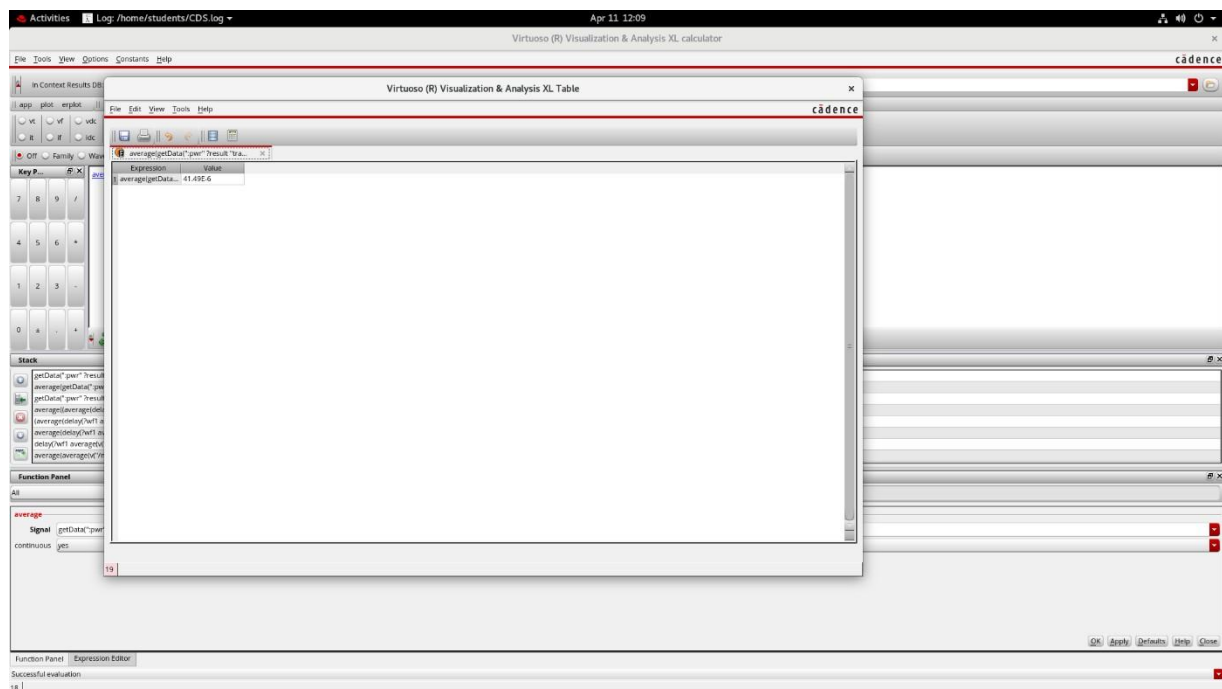
CHAPTER 6

OUTPUT









Comparison between powers of parity generator and parity checker design using static CMOS XOR and transmission gate XOR:

	Static CMOS XOR	Transmission gate XOR
Parity generator	30.09E.6	2.503E.6
Parity Checker	41.49E.6	5.614E.6

CHAPTER 7 CONCLUSION

The 4-bit parity encoder for digital communication systems has been successfully designed and implemented using digital electronics principles. The system effectively achieves its objective of detecting errors by generating a parity bit for a given 4-bit input data. This ensures reliable data transmission by enabling error detection at the receiver end.

By using logic gates, the encoder integrates multiple components into a single functional unit capable of producing either even or odd parity. The implementation of parity logic allows the system to monitor data integrity, ensuring that transmitted information can be verified for correctness. This improves the overall reliability of digital communication systems.

The encoder generates a parity bit based on the input combination, which is then transmitted along with the data. At the receiving end, the parity can be checked to identify whether an error has occurred during transmission. This makes the system simple yet effective for basic error detection.

Furthermore, the design demonstrates how fundamental digital communication concepts can be implemented using combinational logic circuits. The project highlights the importance of error detection techniques in modern communication systems.

Overall, the project is efficient, reliable, and easy to implement. It can be further extended by incorporating advanced error detection and correction techniques such as Hamming code or CRC. This project serves as a practical example of applying digital electronics in communication systems to ensure accurate and secure data transmissio